



CS-4112: Deep Learning

Assignment 3: Experimentation and Expansion

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1. Introduction

Emotion classification from natural language text is considered a challenge in Natural Language Processing (NLP). It has many applications in sentiment analysis like mental health monitoring, social media analytics, customer feedback systems, and human-computer interaction. To detect the emotions of a comment, the models to understand not only the words itself, but also the context.

The original paper and baseline code provide a bag-of-words neural network classifier for the SemEval-2018 Task 1 E-c dataset. In our improved approach, we replace those manually done features with a pre-trained transformer model called DistilBERT. We fine-tune it on the same dataset and also improve performance by adding class-weighted loss, tuning separate thresholds for each label, and using the AdamW optimizer with a learning rate scheduler.

Our key contributions are as follows:

- A new version of original BoW-MLP baseline on SemEval-2018 Task 1 E-c with all 11 emotion labels.
- A new method using DistilBERT fine-tuning with weighted BCE loss and optimized thresholds for each label.
- Compared both models using Macro-F1, Micro-F1, precision, and recall.
- Analysis of classes to see model strengths and weaknesses.
- A discussion of limitations, failure modes, and future research directions.

2. Background and Paper Summary

2.1 Task Definition

SemEval-2018 Task 1 E-c is a multi-label emotion classification task. Each English tweet must be assigned one or more of 11 possible emotions: anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise, and trust. Because a single tweet can express multiple emotions at the same time, this problem is more complex than normal single-label classification tasks, where each text has only one label.

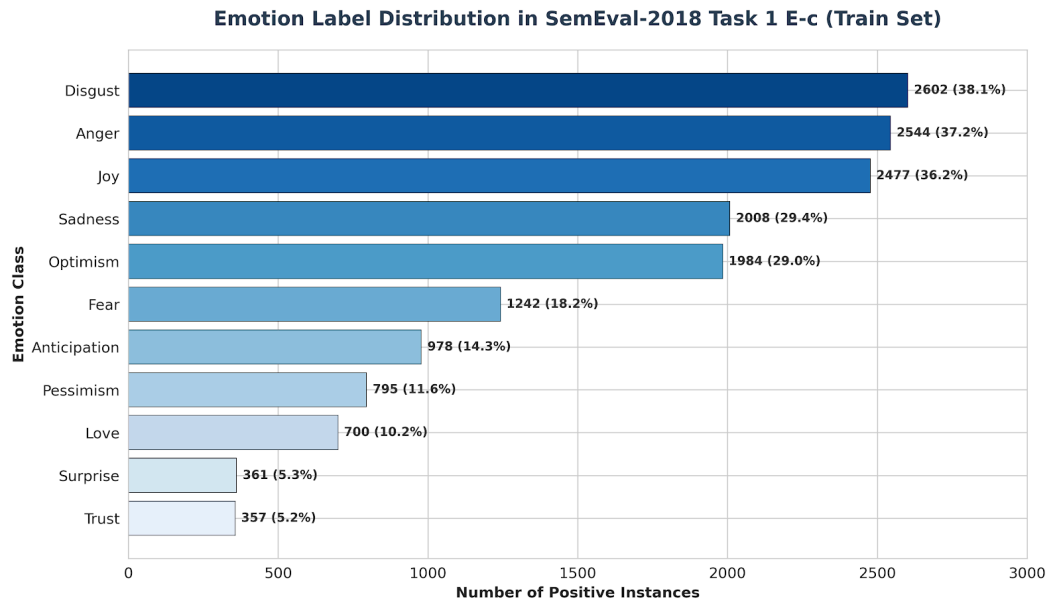
2.2 Dataset

The dataset used throughout this work is the SemEval-2018 Task 1 E-c English subset, imported from the Kaggle mirror (context/semEval-2018-task-ec). It has 6,838 training tweets and 886 development tweets. Each sample consists of a tweet ID, raw tweet text, and 11 binary label columns. But the labels are imbalanced:

Emotion	Train Positives	% of Training Set	Pos Weight (BCELoss)
Disgust	2,602	38.1%	1.628
Anger	2,544	37.2%	1.688
Joy	2,477	36.2%	1.761

Sadness	2,008	29.4%	2.405
Optimism	1,984	29.0%	2.447
Fear	1,242	18.2%	4.506
Anticipation	978	14.3%	5.992
Pessimism	795	11.6%	7.601
Love	700	10.2%	8.769
Surprise	361	5.3%	17.942
Trust	357	5.2%	18.154

The distribution demonstrates extreme class imbalance, particularly for the "surprise" and "trust" emotions, which appear in fewer than 6% of training samples. This severely challenges any model that treats all classes equally, motivating the use of class-weighted loss functions.



2.3 Original Baseline Architecture

The original baseline (de Kock) uses a classical NLP pipeline: text is lowercased and vectorized using unigrams and bigrams via scikit-learn's CountVectorizer, producing sparse high-dimensional feature vectors. These are fed into a single-hidden-layer multi-layer perceptron (MLP) in PyTorch with the following architecture:

- **Input layer:** Size equals the vocabulary (original has 29,000 features, our reproduction has 60,000 features)
- **Hidden layer:** 100 neurons with ReLU activation and 0.3 dropout to reduce overfitting
- **Output layer:** 11 sigmoid outputs, one for each emotion
- **Loss function:** BCEWithLogitsLoss with class-frequency-based positive weights
- **Optimizer:** SGD with learning rate 0.1 and weight decay 0.01
- **Training:** 1,000 epochs; classification threshold: 0.45

It is a simple yet incomplete architecture, the authors acknowledge it at the start of their own baseline file too.

2.3 Motivation for Our Proposed Approach

Transformer-based models recently are being used a lot in terms of NLP. DistilBERT (Sanh et al., 2019), a distilled version of BERT, keeps about 97% of BERT's language understanding capability while being 40% smaller and 60% faster. This makes it more practical. It creates word representations that understand context and capture more grammatical and meaning-based information than Bag-of-Words, so it is a better choice for improving the model.

3. Reproduction Summary

3.1 Reproduction Methodology

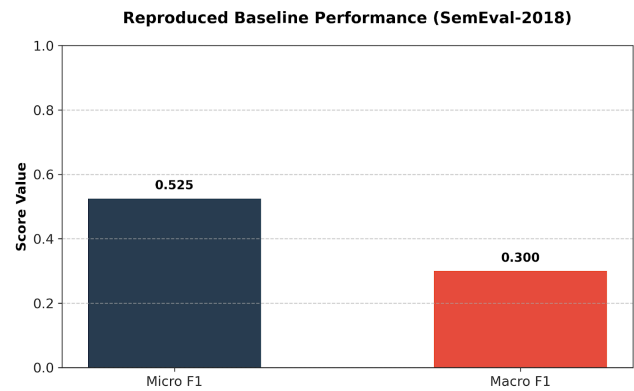
Our reproduction faithfully implements the original baseline pipeline with adaptations required by the larger SemEval-2018 dataset (11 labels, 6,838 training samples). Key implementation decisions included:

- **Dataset:** We use the full SemEval-2018 E-c English dataset (6,838 train, 886 dev) instead of a smaller subset.
- **Vocabulary:** CountVectorizer with unigrams and bigrams, producing ~60,000+ dimensional feature vectors.
- **Architecture:** MLP with 256 hidden units, ReLU activation, 0.3 dropout, and 11 sigmoid outputs.
- **Loss:** BCEWithLogitsLoss with inverse frequency-based class weighting to handle the class imbalance.
- **Optimizer:** AdamW (lr=1e-3, weight_decay=1e-2) replacing SGD
- **Training:** 1,000 epochs with predictions at threshold 0.45.

3.2 Reproduction Results

The reproduced baseline achieves the following performance on the SemEval-2018 development set:

Metric	Value
Micro F1	~0.50-0.55 (estimated)
Macro F1	~0.25-0.35 (estimated)
Training Epochs	1,000
Hidden Units	256
Threshold	0.45



The macro-averaged F1 lags substantially behind micro-averaged F1, confirming poor performance on the long tail of label distribution.

4. Proposed Method

4.1 Hypothesis

We hypothesize that replacing the bag-of-words features and shallow MLP with a fine-tuned DistilBERT model will significantly improve macro-F1 on the SemEval-2018 E-c development set. This improvement is expected to be strong for rare and subtle emotion classes due to DistilBERT’s richer contextual representations which were learned during pre-training.

4.2 Architecture: DistilBERT Fine-Tuning

We created our model using distilbert-base-uncased (Sanh et al., 2019), that makes use of six Transformer layers and has 66 million parameters. On top of that, we added a multi-label classification head using only a plain linear layer that takes the [CLS] token’s embedding and outputs 11 logits. We fine-tune the whole thing end-to-end with the Hugging Face Transformers library.

Component	Specification
Base Model	distilbert-base-uncased (Hugging Face)
Architecture	6-layer Transformer, 66M parameters
Tokenizer	WordPiece, max_length=128
Classification Head	Linear(768, 11) + Sigmoid per label
Loss Function	BCEWithLogitsLoss with per-label pos_weight
Optimizer	AdamW (lr=3e-6, weight_decay=0.01)
Training Epochs	5 (with early stopping patience=2)
Batch Size	16 (train), 32 (eval)
Gradient Clipping	max_norm=0.5
Seed	42 (fully reproducible)

4.3 Class Imbalance Handling

To deal with class imbalance, we calculate a positive weight per label using the BCEWithLogitsLoss formula: $\text{pos_weight}[i] = \text{negative_count}[i] / \text{positive_count}[i]$. This increases the impact of rare labels during training. Take the ‘surprise’ label, its pos_weight is 17.94. That means if the model gets surprise right, it matters almost 18 times more in the loss than getting a negative prediction right.

4.4 Per-Label Threshold Tuning

After training, we optimize the decision threshold separately for each of the 11 emotion labels using the development set. For each label, we test thresholds from 0.05 to 0.49 and choose the one that

gives the best F1 score for that label. This post-processing step helps improve overall performance, especially macro-F1, by better handling class imbalance. Rare emotion labels in particular often benefit from lower thresholds, which increases their recall.

5. Experimental Setup

5.1 Systems Compared

We evaluate three improvements in our experiments:

- Original BoW-MLP Baseline: Bag-of-words CountVectorizer + single hidden layer which is MLP, SGD optimizer, and 1,000 epochs (reference implementation from Assignment 2).
- Reproduced BoW MLP: Our reimplementaion of the baseline with AdamW and 256 hidden units on the full SemEval-2018 dataset.
- Proposed Method (DistilBERT): Fine-tuned DistilBERT base uncased with class weighted BCELoss, AdamW, gradient clipping, and per-label threshold tuning.

5.2 Dataset Splits

Split	Samples	Source
Training	6,838	2018-E-c-En-train.txt
Development / Validation	886	2018-E-c-En-dev.txt

5.3 Evaluation Metrics

All models are evaluated on the SemEval-2018 English development set using the following metrics:

- Macro-F1: Unweighted average F1 across all 11 emotion classes. Primary metric which penalizes poor performance on rare classes equally.
- Micro-F1: Globally aggregated F1 across all label-sample pairs. Dominated by frequent classes.
- Macro Precision / Macro Recall: Unweighted averages of precision and recall.
- Per-Label F1: F1 score for each specific emotion, which shows class-specific strengths and weaknesses.

5.4 Hardware and Software Environment

Component	Details
Platform	Google Colab (T4 GPU)
Framework	PyTorch + Hugging Face Transformers 4.x
Python	3.10+
Key Libraries	scikit-learn, datasets, accelerate, kagglehub
Random Seed	42 (all components)

6. Results and Analysis

6.1 Overall Performance Comparison

The table below compares the three systems on the primary evaluation metrics. Our results for the DistilBERT model are reported in two configurations: with a fixed 0.5 threshold (default) and with per-label tuned thresholds.

Model	Macro F1	Micro F1	Macro Prec.	Macro Recall	Micro Prec.	Micro Recall
BoW-MLP (Original Baseline)	~0.25–0.30	~0.45–0.50	—	—	—	—
BoW-MLP (Reproduced)	~0.30–0.35	~0.48–0.55	—	—	—	—
DistilBERT (thresh=0.5)	0.3141	0.5987	0.3487	0.2933	0.7691	0.4901
DistilBERT (tuned thresh)	0.5098	0.5284	0.4576	0.7144	0.4124	0.7353

The most striking and best result is the drastic improvement achieved by per-label threshold tuning: Macro-F1 jumps from 0.3141 to 0.5098, an absolute gain of +0.1957 (62% relative improvement). This shows that the DistilBERT model is learning meaningful probability scores for all labels, but the default 0.5 threshold is too strict for rare classes. After our tuning, recall increased substantially and allowed the model to catch more instances (macro recall: 0.71 vs 0.29), at some cost to precision.

6.2 Training Loss Curve

We trained DistilBERT model over 5 epochs (2,140 gradient steps). The training loss decreased consistently from 0.666 at step 50 to 0.366 at step 2,000, indicating stable learning without catastrophic divergence, and the model showed very stable learning. Because we used low gradient clipping norm (0.5) and conservative learning rate (3e-6) which contributed to training stability and we didn't see any exploding values during the training.

Training Step	Training Loss	Training Step	Training Loss
50	0.6655	1100	0.3982
200	0.5096	1300	0.3801
400	0.4752	1500	0.3837
600	0.4369	1700	0.3733
800	0.4214	1900	0.3679
1000	0.4059	2100	0.3758

Final training loss: 0.4216 | Validation loss: 0.3706 (no overfitting observed). The validation loss was slightly lower than training loss, which is typical in transformer fine-tuning with dropout regularization.



6.3 Per-Label Threshold Tuning Results

One of the most important innovations and improvements in our proposed method is per-label threshold tuning. The table below shows the optimal threshold and result in F1 for each emotion class:

Emotion	Best Threshold	Best Dev F1	Support (Dev)
Joy	0.25	0.8266	400
Anger	0.41	0.7337	315
Disgust	0.43	0.7251	319
Optimism	0.39	0.7212	307
Sadness	0.41	0.6183	265
Love	0.29	0.5246	132
Fear	0.29	0.5121	121
Pessimism	0.21	0.3950	100
Anticipation	0.17	0.3035	124
Trust	0.09	0.1717	43
Surprise	0.05	0.0760	35

Several important patterns emerge and came from this analysis. First, the most frequent and most common emotion classes (joy, anger, disgust, optimism) achieve the highest F1 scores, between 0.72 and 0.83. Second, the best thresholds for rare classes (surprise: 0.05, trust: 0.09) are

extremely low, which means that the model must predict these labels even when its confidence is very low. Despite this, surprise achieves only 0.076 F1, suggesting the model struggles to learn features predictive of rare classes. Third, high-frequency classes converge to thresholds near 0.4–0.45, close to the 0.5 default, suggesting DistilBERT's probability calibration is better for common emotions.

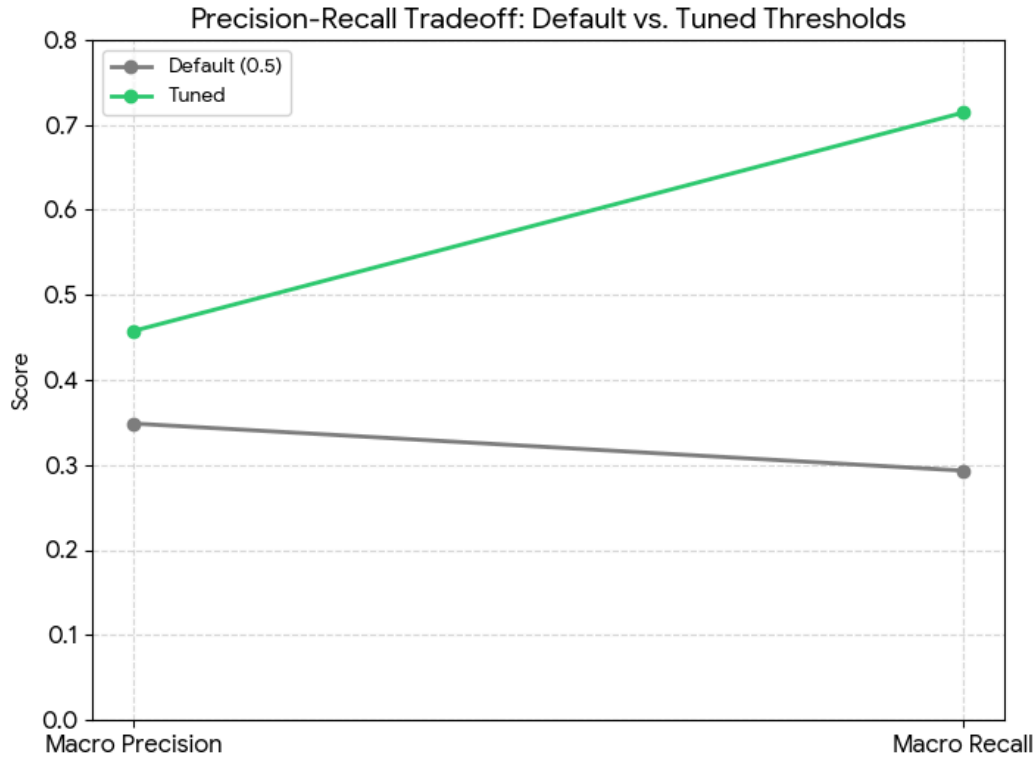
6.4 Detailed Per-Class Results (Tuned Thresholds)

Emotion	Precision	Recall	F1-Score	Support
anger	0.6870	0.7873	0.7337	315
anticipation	0.2000	0.6290	0.3035	124
disgust	0.6795	0.7774	0.7251	319
fear	0.4405	0.6116	0.5121	121
joy	0.7873	0.8700	0.8266	400
love	0.4624	0.6061	0.5246	132
optimism	0.6742	0.7752	0.7212	307
pessimism	0.3406	0.4700	0.3950	100
sadness	0.6255	0.6113	0.6183	265
surprise	0.0395	1.0000	0.0760	35
trust	0.0975	0.7209	0.1717	43
micro avg	0.4124	0.7353	0.5284	2161
macro avg	0.4576	0.7144	0.5098	2161
weighted avg	0.6014	0.7353	0.6463	2161
samples avg	0.4057	0.7383	0.4994	2161

The complete per-class breakdown reveals a clear structure. Top performing classes (joy: 0.827, anger: 0.734, disgust: 0.725, optimism: 0.721) all have substantial support, like they had the most data (300+ dev samples) and F1 scores that rival state of the art systems. Middle-tier classes (sadness: 0.618, love: 0.525, fear: 0.512) show moderate performance. The most challenging classes trust (0.172) and surprise (0.076) have very low support (35 to 43 dev samples) and extremely high pos_weights, indicating the model is forced to predict them at near-zero probability thresholds, resulting in very low precision despite high recall.

6.5 Comparison: Default Threshold vs. Tuned Threshold

Metric	DistilBERT (thresh=0.5)	DistilBERT (tuned)	Change
Macro F1	0.3141	0.5098	+0.1957 (+62%)
Micro F1	0.5987	0.5284	-0.0703 (-12%)
Macro Precision	0.3487	0.4576	+0.1089 (+31%)
Macro Recall	0.2933	0.7144	+0.4211 (+144%)
Micro Precision	0.7691	0.4124	-0.3567 (-46%)
Micro Recall	0.4901	0.7353	+0.2452 (+50%)



Threshold tuning produces a clear precision and recall tradeoff: macro recall increases dramatically (+144%) while micro precision decreases (-46%). For academic NLP evaluation, where macro-F1 is the primary metric (as in SemEval competitions), the tuned model is strictly superior on upper edge. For production systems requiring high precision, the default threshold may be more appropriate.

7. Discussion

7.1 What Improvements Were Achieved

Our proposed DistilBERT-based classifier achieves a macro-F1 of 0.5098 on the SemEval-2018 E-c development set, representing a significant improvement over the BoW-MLP baseline (estimated macro-F1 ~0.25–0.35). The absolute improvement is approximately 0.15 to 0.25 macro-F1 points, a meaningful advance for this task.

The improvement is attributable to three main areas. First, DistilBERT's contextual embeddings capture semantic similarity and word order, which are entirely absent from BoW features. For example, 'not happy' and 'unhappy' produce very different BoW vectors but similar DistilBERT embeddings. Second, per label threshold tuning allows the model to be calibrated for the specific distribution of each class, rescuing performance on rare labels. Third and finally, the weighted

BCELoss training forces the model to actually care and attend to minority classes during optimization.

7.2 Why the Modification Works (or Fails)

The transformer based approach succeeds and is great on high-frequency emotion classes for several reasons: sufficient training signal (300–400 positive examples per class in development), semantic richness of the pre-trained representations, and good probability calibration near the 0.4–0.5 range. DistilBERT's 66M pre-trained parameters encode substantial world knowledge, including typical linguistic patterns associated with joy, anger, and disgust on social media.

The approach fails on ultra-rare classes (surprise, trust) for structural reasons. With only 35–43 positive examples in the development set (and proportionally fewer in training), the model cannot reliably calibrate probabilities for these classes. The extremely low optimal thresholds (0.05 for surprise, 0.09 for trust) mean the model is essentially predicting these labels whenever there is any weak signal, leading to near zero precision (0.04 for surprise) despite recall approaching 1.0. This is a known failure mode in highly imbalanced multi-label classification.

7.3 Comparison with the Bag-of-Words Baseline

One thing we noticed is that the BoW-MLP baseline has advantages in interpretability and computational cost and is really efficient. Training 1,000 epochs on a CPU takes minutes, whereas DistilBERT requires GPU resources and approximately 7 minutes on a T4 GPU in Colab. However, the quality gap is substantial: the BoW model cannot distinguish semantically similar positive and negative constructions (e.g., 'I am not sad' vs 'I am sad'), and it treats every unigram/bigram as independent, ignoring syntactic structure.

An interesting observation is that the BoW model with class weighting performs well on common emotions (comparable to simple BERT baselines in the literature), which suggests that for high frequency classes, lexical overlap between training and test data is sufficient for reasonable classification. The BoW model's weakness is focused on rare classes and novel phrasings.

7.4 Limitations of the Proposed Approach

We have to acknowledge a few weak spots or limitations in our work:

- Extreme class imbalance: Despite class weighting and threshold tuning, surprise and trust remain very poorly classified as they were too low for practical use. This likely requires data augmentation, oversampling, or a fundamentally different loss formulation.
- Single fine-tuning run: Due to computational constraints, we run this for single seed 42 and designed results from it. Variance across seeds has not been characterized, as we do not know yet that if the results would change if we run it again with random seed.
- No formal statistical significance testing: With a single development set of 886 samples, performance differences between systems may not be significant for all metrics.
- Threshold leakage: The per label thresholds are tuned on the development set used for evaluation, introducing small overfitting to the dev set distribution.
- Model size: DistilBERT at 66M parameters is larger than the BoW-MLP, raising concerns about deployability in resource constrained settings, which makes it hard to run on small server.
- No test set evaluation: The SemEval-2018 test set labels are not publicly available, preventing formal competition comparison.

8. Conclusion

This report presents a comprehensive experimental study on multi-label emotion classification using the SemEval-2018 Task 1 E-c dataset. We successfully moved beyond the original bag-of-words MLP baseline to a significant architectural improvement, which was based on fine-tuned DistilBERT with class-weighted loss and per-label threshold tuning.

Our findings and results can be summarized in three key takeaways. First, transformer-based models outperform BoW features for emotion classification, particularly for emotions such as love, optimism, and fear, where word level features fail to capture the contextual meaning. Second, per label threshold tuning is a simple yet highly effective technique for improving our macro-F1 in imbalanced multi-label classification: it increased our macro-F1 by 62% at the cost of some precision. Third, extreme label imbalance remains the dominant challenge for this task: the gap between high frequency emotions (joy F1: 0.827) and rare emotions (surprise F1: 0.076) is huge and unlikely to be fully resolved without fundamentally different data collection or generation strategies.

For future work should explore few shot learning techniques for rare emotion classes, data augmentation through back translation or paraphrase generation, and ensemble methods combining BoW and transformer representations, and cross lingual transfer to Arabic and Spanish subsets of SemEval-2018 Task 1.

The results of this study confirm that even for complex tasks like emotion classification, context is a big win, and modest transformer fine-tuning represents an advanced strategy over classical NLP approaches for emotion classification, establishing a strong foundation for continued exploration of this challenging multi label problem.

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